LESSON 2: BOUNDED LINEAR OPERATORS ON BANACH SPACES
by F. Colombo and A. De Martino

In this lesson, we define the notion of bounded linear operator on a Banach space, present its main properties, and provide several examples. We will then focus on the different types of convergence that can be defined in the space of all bounded linear operators. Finally we will introduce the Neumann series, which is crucial when examining resolvent operators.

Definition 0.1 (Definition of linear operator)**.** *Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be Banach spaces over the same field $\mathbb{F}$ (that can be $\mathbb{R}$ or $\mathbb{C}$). A linear operator $A : X \to Y$ is a function that satisfies the following linearity conditions:*

(1) ***Additivity**: For all $x, y \in X$,*

$$A(x + y) = A(x) + A(y).$$

(2) ***Homogeneity**: For all $x \in X$ and all scalars $\alpha \in \mathbb{F}$*

$$A(\alpha x) = \alpha A(x).$$

Definition 0.2 (Bounded linear operator)**.** *Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be Banach spaces over the same field $\mathbb{F}$. A linear operator $A : X \to Y$ is said to be bounded if there exists a constant $C \geq 0$ such that for all $x \in X$,*

$$\|A(x)\|_Y \leq C\|x\|_X.$$

The set of all bounded linear operators $A : X \to Y$ will be denoted by $\mathcal{B}(X, Y)$. The smallest such constant C is called the operator norm of A and is denoted by $\|A\|$.

Theorem 0.3. *The norm of A defined as*

$$\|A\| = \sup_{x \neq 0} \frac{\|A(x)\|_Y}{\|x\|_X}$$

can be written as:

$$\sup_{x \neq 0} \frac{\|A(x)\|_Y}{\|x\|_X} = \sup_{\|x\|_X \leq 1} \|A(x)\|_Y = \sup_{\|x\|_X = 1} \|A(x)\|_Y.$$

We will now define the identity operator and the zero operator, which will be used later in the discussion:

- **Identity Operator**: The identity operator I_X on X is defined by

  $$I_X(x) = x$$

  for all $x \in X$. It is an element of $\mathcal{B}(X, X)$ with $\|I_X\| = 1$.
- **Zero Operator**: The zero operator 0_{oper} is defined by

  $$0_{oper}(x) = 0$$

  for all $x \in X$. It is an element of $\mathcal{B}(X, X)$ with $\|0_{oper}\| = 0$.

Remark 0.4. *In the following, when we consider an operator A acting between Banach spaces $(X, |\cdot|_X)$ and $(Y, |\cdot|_Y)$, we always assume that both spaces are defined over the same field $\mathbb{F}$. For the sake of brevity, we will often omit the phrase: "over the same field $\mathbb{F}$".*

Remark 0.5. *Recall the definition of an Algebra over a field $\mathbb{F}$.*

Definition 0.6. *An **algebra** over a field $\mathbb{F}$ is a vector space $\mathcal{A}$ equipped with a bilinear product $\cdot : \mathcal{A} \times \mathcal{A} \to \mathcal{A}$ such that for each $x, y, z \in \mathcal{A}$ and $\alpha \in \mathbb{F}$:*

- ***Associativity**: $x \cdot (y \cdot z) = (x \cdot y) \cdot z$*
- ***Distributivity**: $x \cdot (y + z) = x \cdot y + x \cdot z$ and $(x + y) \cdot z = x \cdot z + y \cdot z$*
- ***Scalar Multiplication**: $\alpha(x \cdot y) = (\alpha x) \cdot y = x \cdot (\alpha y)$*

*An algebra is said to have a **unit** if there exists an element $1 \in \mathcal{A}$ such that for every $x \in \mathcal{A}$:*

$$x \cdot 1 = 1 \cdot x = x.$$

In this case, $\mathcal{A}$ is often called an algebra with unit or a unital algebra.

Theorem 0.7 (Algebra of Bounded Linear Operators)**.** *Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be Banach spaces. Let $\mathcal{B}(X, Y)$ denote the space of all bounded linear operators from X to Y. The space $\mathcal{B}(X, Y)$ forms an algebra under the following operations:*

Remark 0.8 (Algebraic Operations)**.** *(Addition and multiplication by scalar make $\mathcal{B}(X, Y)$ a vector space)*

- ***Addition**: If $A, B \in \mathcal{B}(X, Y)$, then their sum $A + B$ is defined by*

  $$(A + B)(x) = A(x) + B(x)$$

  for all $x \in X$. The sum $A + B$ is also a bounded linear operator and

  $$\|A + B\| \leq \|A\| + \|B\|.$$

- ***Scalar Multiplication**: For a scalar $\alpha \in \mathbb{F}$ and $A \in \mathcal{B}(X, Y)$, the scalar multiple αA is defined by*

  $$(\alpha A)(x) = \alpha A(x)$$

  for all $x \in X$. The operator αA is also a bounded linear operator and

  $$\|\alpha A\| = |\alpha| \cdot \|A\|.$$

- **Composition (Operator Multiplication)**: If $A \in \mathcal{B}(Y, Z)$, and $B \in \mathcal{B}(X, Y)$ then the composition $A \cdot B$ is defined by

  $$(A \cdot B)(x) = A(B(x)) \quad \text{(We will often write } AB \text{ instead of } A \cdot B)$$

  for all $x \in X$. The composition AB is a bounded linear operator from X to Z and

  $$\|AB\| \leq \|A\|\|B\|.$$

Remark 0.9 (Properties of Algebraic Operations)**.** *Considering the appropriate spaces on which the bounded linear operators A, B and C act we also have*

- ***Associativity***: $A \cdot (B \cdot C) = (A \cdot B) \cdot C$
- ***Distributivity***:
$$A \cdot (B + C) = A \cdot B + A \cdot C$$
and
$$(A + B) \cdot C = A \cdot C + B \cdot C$$
- ***Scalar Multiplication***:
$$\alpha(A \cdot B) = (\alpha A) \cdot B = A \cdot (\alpha B)$$

Proposition 0.10. *Properties of $\mathcal{B}(X, Y)$*

- *The set $\mathcal{B}(X, Y)$ forms a vector space under addition and scalar multiplication.*
- *The set $\mathcal{B}(X, Y)$ is a normed algebra under the operator norm.*
- *$\mathcal{B}(X, Y)$ is a Banach space if Y is a Banach spaces, meaning $\mathcal{B}(X, Y)$ is complete with respect to the operator norm.*

0.1. Examples of Bounded Linear Operators.

Example 0.11 (Identity Operator)**.** *Let X be a Banach space. The identity operator $I_X : X \to X$ defined by $I_X(x) = x$ for all $x \in X$ is bounded. In fact, we have for all $x \in X$,*
$$\|I_X(x)\|_X = \|x\|_X$$
and moreover
$$\|I_X\|_{\mathcal{B}(X,X)} = \sup_{x \neq 0} \frac{\|I(x)\|_X}{\|x\|_X} = \sup_{x \neq 0} \frac{\|x\|_X}{\|x\|_X} = 1.$$

Example 0.12 (Scalar Multiplication Operator)**.** *Let X be a real Banach space and $\alpha \in \mathbb{R}$. The scalar multiplication operator $A_\alpha : X \to X$ defined by $A_\alpha(x) := \alpha x$ for all $x \in X$ is bounded. For all $x \in X$,*
$$\|A_\alpha(x)\|_X = \|\alpha x\|_X = |\alpha| \|x\|_X.$$
Hence, the scalar multiplication operator A_α has the norm $\|A_\alpha\| = |\alpha|$.

Example 0.13 (Shift Operator on ℓ^p Spaces)**.** *Let $1 \leq p < \infty$ and consider the space ℓ^p. The shift operator $S : \ell^p \to \ell^p$ defined by*
$$A((x_1, x_2, x_3, \ldots)) = (0, x_1, x_2, x_3, \ldots)$$
is bounded. Let $x = (x_1, x_2, x_3, \ldots) \in \ell^p$. Then,
$$\begin{aligned} \|A(x)\|_{\ell^p} &= (|0|^p + |x_1|^p + |x_2|^p + |x_3|^p + \cdots)^{1/p} \\ &= \left(\sum_{n=1}^{\infty} |x_n|^p \right)^{1/p} = \|x\|_{\ell^p}. \end{aligned}$$
Thus, the shift operator A is bounded with $\|A\| = 1$.

Example 0.14 (Integral Operator on continuous functions)**.** *Let $X = C([a,b])$ be the space of continuous functions on the interval $[a,b]$ with the supremum norm. Define the integral operator $A : C([a,b]) \to C([a,b])$ by*

$$(Af)(x) = \int_a^x f(t)\,dt, \quad x \in [a,b].$$

Let $f \in C([a,b])$. Then,

$$\|Af\|_\infty = \sup_{x\in[a,b]} \left| \int_a^x f(t)\,dt \right| \leq \sup_{x\in[a,b]} \int_a^x |f(t)|\,dt \leq \sup_{x\in[a,b]} (x-a)\|f\|_\infty.$$

Since $x \in [a,b]$, we have $x - a \leq b - a$. Therefore,

$$\|Af\|_\infty \leq (b-a)\|f\|_\infty.$$

Hence, the integral operator A is bounded with $\|A\| \leq b - a$.

Example 0.15 (**Not all linear operators are bounded: An Unbounded Operator**)**.** *The simplest example of an unbounded operator is the differentiation operator*

$$D_x : C^1([a,b]) \to C^0([a,b])$$

defined on the Banach space $C^1([a,b])$ of all continuously differentiable functions (with the natural norm) into the space $C^0([a,b])$ of all continuous functions. Observe that

$$f_n(x) := \sin(nx) \ \text{ is such that } \ \|f_n\|_{C^0([a,b])} = \max_{x\in[a,b]} |f_n(x)| = 1$$

but

$$D_x f_n(x) = n\cos(nx) \quad \text{so} \quad \|D_x f_n\|_{C^0([a,b])} = n$$

which says that there are no positive constants C such that

$$\|D_x f_n\|_{C^0([a,b])} \leq C\|f_n\|_{C^1([a,b])}$$

So D_x is an unbounded operator because it takes bounded sets into unbounded ones.

Definition 0.16 (Adjoint or dual Operator)**.** *Let $\mathcal{B}(Y) := \mathcal{B}(Y,Y)$. We also set $Y^* = \mathcal{B}(Y,F)$, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, and we write*

$$y^*(y) = (y, y^*)_{Y,Y^*} = (y, y^*)$$

for $y \in Y$ and $y^* \in Y^*$.

- *Let $A \in \mathcal{B}(X,Y)$, then the adjoint or dual operator A^* is defined on Y^* (the dual space of Y) by*

  $$(A(f), g) = (f, A^*(g))$$

  for all $g \in Y^*$ and $f \in X$. The adjoint operator A^* is an element of $\mathcal{B}(Y^*, X^*)$ and

  $$\|A^*\| = \|A\|.$$

Remark 0.17. *Observe that working in Banach spaces the symbol $(y, y^*)_{Y,Y^*}$ denotes the duality: y^* is a bounded linear functional in Y^* acting on the element $y \in Y$.*

Remark 0.18 (Examples of Adjoint)**.** *We consider the following examples:*

(1) *Adjoint of a Matrix Operator*
(2) *Adjoint of an Integral Operator*

Example 0.19 (Adjoint of a Matrix Operator)**.** *Consider $X = Y = \mathbb{R}^n$ with the standard inner product, and let $A : \mathbb{R}^n \to \mathbb{R}^n$ be represented by a matrix M_A. The adjoint operator A^* corresponds to the transpose of the matrix M_A. Let M be a matrix in $\mathbb{R}^{n\times n}$. The adjoint operator A^* of the matrix operator A is given by*

$$A^* = M_A^\tau,$$

where M_A^τ denotes the transpose of M_A. If A is represented by

$$A(\mathbf{x}) = M_A\mathbf{x},$$

then the adjoint operator A^* acts on $\mathbf{y} \in \mathbb{R}^n$ as

$$\langle M_A\mathbf{x}, \mathbf{y}\rangle = \langle \mathbf{x}, M_A^\tau\mathbf{y}\rangle.$$

Example 0.20 (Adjoint of an Integral Operator)**.** *Let $X = L^2([a, b])$ and $Y = L^2([a, b])$ for a compact set $[a, b]$ in $\mathbb{R}$. Consider an integral operator A defined by*

$$(Af)(t) := \int_a^b K(t, s)f(s)\, ds,$$

where $K(t, s)$ is a kernel function such that $K \in L^2([a, b] \times [a, b])$. The adjoint operator A^* is

$$(A^*g)(t) = \int_a^b K(t, s)g(s)\, ds.$$

This directly comes from the definition. In fact, consider the standard inner product in $L^2([a, b])$,

$$(Af, g) = \int_a^b \Big(\int_a^b K(t, s)f(s)\, ds\Big)g(t)dt =$$

$$= \int_a^b \int_a^b K(t, s)f(s)\, g(t)dsdt = \int_a^b f(t)\Big(\int_a^b K(t, s)g(s)\, ds\Big)ds = (f, A^*g),$$

where we have exchanged the roles of s and t. If for example $K(x, t) = e^{x+t}$, then the adjoint operator A^* is given by

$$(A^*g)(t) = \int_a^b e^{x+t}g(x)\, dx.$$

1. Definition of Continuity for Linear Operators

The set of bounded linear operators is equal to the set of continuous linear operators.

Definition 1.1 (Continuous Linear Operator)**.** *Let Banach spaces X and Y be Banach spaces. A linear operator $A : X \to Y$ is said to be* continuous *if for every sequence (x_n) in X that converges to some $x \in X$, the sequence $(A(x_n))$ converges to $A(x)$ in Y, which means:*

$$\text{If } x_n \to x \text{ in } X, \text{ then } A(x_n) \to A(x) \text{ in } Y.$$

Theorem 1.2. *A linear operator $A : X \to Y$ between Banach spaces X and Y is continuous if and only if it is bounded.*

2. Convergence in the Space of Bounded Linear Operators

There are several notions of convergence that can be defined in $\mathcal{B}(X, Y)$:

Definition 2.1 (Converge in Norm)**.** *A sequence of bounded linear operators $\{A_n\} \subset \mathcal{B}(X, Y)$ is said to* converge in norm *to a bounded linear operator $A \in \mathcal{B}(X, Y)$ if*

$$\|A_n - A\| \to 0 \quad \text{as} \quad n \to \infty.$$

Here, $\|\cdot\|$ denotes the operator norm.

Example 2.2 (Converge in Norm)**.** *Let $A_n, A \in \mathcal{B}(\ell^2)$ be defined by*

$$A_n(x) = \left(\frac{x_1}{n+1}, \frac{x_2}{n+1}, \dots\right) \quad \text{and} \quad A(x) = 0.$$

where take $x = (x_1, \dots, x_n, \dots) \in \ell^2$. So we have

$$\|A_n(x) - A(x)\|_{\ell^2}^2 = \|A_n(x) - 0\|_{\ell^2}^2 = \sum_{j=1}^{\infty} \left(\frac{x_j}{n+1}\right)^2 = \frac{1}{(n+1)^2}\|x\|_{\ell^2}^2.$$

so we have

$$\|A_n - A\|_{\mathcal{B}(\ell^2)} = \sup_{\|x\|_{\ell^2}^2 \neq 0} \frac{\|A_n(x) - A(x)\|_{\ell^2}}{\|x\|_{\ell^2}} = \frac{1}{n+1} \quad \to 0$$

which means A_n converges to A in norm.

Definition 2.3 (Strong Convergence)**.** *A sequence of bounded linear operators $\{A_n\} \subset \mathcal{B}(X, Y)$ is said to* converge strongly *to a bounded linear operator $A \in \mathcal{B}(X, Y)$ if*

$$A_n x \to Ax \quad \text{for all} \quad x \in X.$$

Example 2.4 (Strong Convergence)**.** *Consider, in*

$$X = \ell^2 = \{x = (x_1, \dots, x_n, \dots) \ \sum |x_n|^2 < \infty\}.$$

the operators $A_n(x) = (0, \dots, 0, x_n, 0, \dots)$ and $A = 0_{operator}$.

For all $x \in X$, we have strong convergence:

$$\|A_n(x) - A(x)\|_{\ell^2} = \|(0, \ldots, 0, x_n, 0, \ldots)\|_{\ell^2} = |x_n| \to 0, \quad \textit{since} \quad \sum |x_n|^2 < \infty.$$

But we do not have converge in norm:

$$\|A_n - A\|_{\mathcal{B}(\ell^2)} = \sup_{\|x\|_{\ell^2} \neq 0} \frac{\|A_n x - Ax\|_{\ell^2}}{\|x\|_{\ell^2}} = \|A_n\|_{\mathcal{B}(\ell^2)} = 1 \not\to 0 \ \textit{as} \quad n \to \infty$$

when we take $x = (0, \ldots, 0, x_n, 0, \ldots)$ so it does not converge in norm.

Example 2.5. *Projections in $X = \ell^2$. Let*

$$P_n(x_1, x_2, x_3, \ldots) = \underbrace{(0, \ldots, 0}_{n \ \textit{places}}, x_{n+1}, x_{n+2}, x_{n+3}, \ldots)$$

be the projection on the orthogonal complement of the first n components. Then for each fixed $x \in \ell^2$,

$$\|P_n x\|_H^2 = \sum_{i=n+1}^{\infty} |x_i|^2$$

converges to zero as $n \to \infty$. So $\lim_{n\to\infty} P_n x = 0$ and P_n converges strongly to 0 as $n \to \infty$. But, for any $n > m$,

$$(P_m - P_n)(x_1, x_2, x_3, \ldots) = \underbrace{(0, \ldots, 0}_{m \ \textit{places}}, x_{m+1}, x_{m+2}, \ldots, x_n, 0, \ldots)$$

so that there is a vector $x \in \ell^2$ with $(P_m - P_n)x = x$.

Consequently, $\|P_m - P_n\| = 1$, so is not Cauchy, and does not converge in norm.

Definition 2.6. *A series $\sum_{n=0}^{\infty} A_n$ of bounded linear operators is said to be* convergent *the operator norm if there exists a bounded linear operator $A : X \to Y$ such that*

$$\left\| A - \sum_{n=0}^{N} A_n \right\| \to 0 \quad \textit{as} \quad N \to \infty.$$

3. Neumann Series of Bounded Linear Operators

Let X be a Banach space and $A \in \mathcal{B}(X)$ be a bounded linear operator on X. We want to find conditions such that the series associated with A is given by

$$\sum_{n=0}^{\infty} A^n = I + A + A^2 + A^3 + \cdots,$$

where I is the identity operator on X, is convergent in the norm.

Definition 3.1. *The inverse A^{-1} of a bounded linear operator A is an operator such that*

$$AA^{-1} = A^{-1}A = I.$$

We will discuss with more details the inverse of an operator in the next lesson.

Theorem 3.2. *Let $A \in \mathcal{B}(X)$ be a bounded linear operator on a Banach space X. If $\|A\| < 1$, then the Neumann series $\sum_{n=0}^{\infty} A^n$ converges in the operator norm and its sum is given by*

$$(I - A)^{-1} = \sum_{n=0}^{\infty} A^n.$$

Proof of the convergence of the Neumann series. Let

$$S_N = \sum_{n=0}^{N} A^n.$$

We will show that $\{S_N\}$ is a Cauchy sequence in $\mathcal{B}(X)$. This implies that $\{S_N\}$ converges to some $S \in \mathcal{B}(X)$ because X is a Banach space and so $\mathcal{B}(X)$ is a Banach spaces as well.

For $m > n$, we have

$$S_m - S_n = \sum_{k=n+1}^{m} A^k.$$

Then,

$$\|S_m - S_n\| = \left\| \sum_{k=n+1}^{m} A^k \right\| \leq \sum_{k=n+1}^{m} \|A^k\| = \sum_{k=n+1}^{m} \|A\|^k,$$

where we have used the properties of the operator norm

$$\|A + B\| \leq \|A + \|B\| \quad \text{and} \quad \|AB\| \leq \|A\|\|B\|.$$

Since $\|A\| < 1$, the geometric series converges $\sum_{j=0}^{\infty} \|A\|^j < \infty$. Specifically,

$$\sum_{k=n+1}^{m} \|A\|^k = \|A\|^{n+1} \sum_{j=0}^{m-n-1} \|A\|^j \leq \|A\|^{n+1} \sum_{j=0}^{\infty} \|A\|^j = \frac{\|A\|^{n+1}}{1 - \|A\|}.$$

Given $\epsilon > 0$, choose N such that

$$\frac{\|A\|^{N+1}}{1 - \|A\|} < \epsilon.$$

For $m, n > N$, it follows that

$$\|S_m - S_n\| < \epsilon,$$

proving that $\{S_N\}$ is a Cauchy sequence in $\mathcal{B}(X)$.

Since $\mathcal{B}(X)$ is a Banach space, $\{S_N\}$ converges to some $S \in \mathcal{B}(X)$.

Next, we show that $S = (I - A)^{-1}$. Consider

$$(I - A)S = (I - A)\sum_{n=0}^{\infty} A^n = \sum_{n=0}^{\infty} A^n - \sum_{n=0}^{\infty} A^{n+1}.$$

The right-hand side is a telescoping series:

$$\sum_{n=0}^{\infty} A^n - \sum_{n=0}^{\infty} A^{n+1} = (I + A + A^2 + \ldots) - (A + A^2 + \ldots) = I$$

Thus,

$$(I - A)S = I,$$

Similarly we can show that $S(I - A) = I$ which implies $S = (I - A)^{-1}$. Therefore,

$$(I - A)^{-1} = \sum_{n=0}^{\infty} A^n,$$

proving the convergence of the Neumann series when $\|A\| < 1$.

□

Corollary 3.3. *If $A \in \mathcal{B}(X)$ and $\|A\| < 1$, then the operator $(I - A)^{-1}$ is well-defined, linear, bounded and given by*

$$(I - A)^{-1} = \sum_{n=0}^{\infty} A^n$$

and the estimate

$$\|(I - A)^{-1}\| \leq \frac{1}{1 - \|A\|}$$

holds.

Proof. We just have to prove the estimate. By the geometric series

$$\sum_{n=0}^{\infty} z^n = \frac{1}{1 - z}, \qquad |z| < 1$$

and the property $\|A^n\| \leq \|A\|^n$ we have

$$\|(I - A)^{-1}\| = \|\sum_{n=0}^{\infty} A^n\| \leq \sum_{n=0}^{\infty} \|A^n\| \leq \sum_{n=0}^{\infty} \|A\|^n = \frac{1}{1 - \|A\|}.$$

□

Example 3.4. *Let $X = \ell^2$ for any $x = (x_1, x_2, ...) \in \ell^2$ (that is, $\sum_{k=1}^{\infty} |x_k|^2 < \infty$), and define $A : \ell^2 \to \ell^2$ by*

$$A(x) = \left(0, \frac{x_1}{2}, \frac{x_2}{2}, \frac{x_3}{2}, \ldots\right).$$

Determine explicitly the operator $(I-A)^{-1}$. *Since*

$$\|A(x)\|_{\ell^2}^2 = \sum_{k=1}^{\infty} \frac{x_k^2}{4} = \frac{1}{4}\|x\|_{\ell^2}^2$$

we can write

$$\|A(x)\|_{\ell^2} = \frac{1}{2}\|x\|_{\ell^2}$$

which implies that

$$\|A\| = \sup_{x\neq 0} \frac{\|A(x)\|_{\ell^2}}{\|x\|_{\ell^2}} = \frac{1}{2}$$

so $\|A\| < 1$. *By the Neumann series*

$$(I-A)^{-1} = \sum_{n=0}^{\infty} A^n$$

we have

$$\sum_{n=0}^{\infty} A^n(x) = (x_1, x_2, x_3, \ldots) + \left(0, \frac{x_1}{2}, \frac{x_2}{2}, \frac{x_3}{2}, \ldots\right) + \left(0, 0, \frac{x_1}{4}, \frac{x_2}{4}, \ldots\right) + \cdots,$$

where $A^0 = I$. *The series converges in* ℓ^2 *norm and we also have*

$$\|(I-A)^{-1}\| \leq \frac{1}{1-\|A\|} = \frac{1}{1-\frac{1}{2}} = 2.$$

Remark 3.5. *An observation which is useful in quantum mechanics.*

Example 3.6. *Let* H *be a Hilbert space, and let* $A, B \in \mathcal{B}(H)$.
Show that the relation $AB - BA = I$ *cannot not hold.*

- *By induction, if* $AB - BA = I$, *then we have*

$$AB^n - B^nA = nB^{n-1} \tag{3.1}$$

  for all positive integers $n = 1, 2, 3, \ldots$.
- *In fact multiplying the relation* $AB = BA + I$ *by* B *on the right we get*

$$AB^2 = B(AB) + B$$

  and replacing the same relation we obtain

$$AB^2 = B(AB) + B = AB^2 = B(BA + I) + B = B^2A + 2B$$

  so

$$AB^2 - B^2A = 2B.$$

  As a further step, in order to understand the procedure, multiply the relation

$$AB^2 = B^2A + 2B$$

  by B *on the right we have*

$$AB^2B = (B^2A + 2B)B$$

so that now we proceed as above and get

$$AB^3 = B^2(AB) + 2B^2 = B^2(BA + I) + 2B^2 = B^3A + 3B^2$$

and so on. From the formula (3.1), written as (since $n = 1, 2, 3, ...$)

$$nB^{n-1} = AB^n - B^nA = AB^{n-1}B - B^{n-1}BA$$

taking the norm

$$\|nB^{n-1}\| = \|AB^{n-1}B - B^{n-1}BA\| \leq \|AB^{n-1}B\| + \|B^{n-1}BA\| \leq 2\|A\|\|B^{n-1}\|\|B\|$$

we finally obtain that

$$2\|A\|\|B\| \geq n$$

for every n.

- *This leads to the fact that $\|A\|$ or $\|B\|$ or both cannot be bounded.*

Example 3.7. *The above fact has implication in quantum mechanics.*

- *The position operator $\hat{x}$ defined by $\hat{x}\psi(x) = x\psi(x)$*
- *and the momentum operator $\hat{p}$ defined by $\hat{p} = -i\hbar\frac{d}{dx}$*
- *have a fundamental relationship described by the canonical commutation relation. This relation is central to the Heisenberg uncertainty principle and is given by:*

$$[\hat{x}, \hat{p}] = \hat{x}\hat{p} - \hat{p}\hat{x} = i\hbar$$

  where:
  - *$[\hat{x}, \hat{p}]$ is the commutator of the position and momentum operators.*
  - *i is the imaginary unit.*
  - *$\hbar$ (h-bar) is the reduced Planck's constant, $\hbar = \frac{h}{2\pi}$.*
- *This as implication on the momentum al position operators that do not commute. They cannot be bounded operators.*